

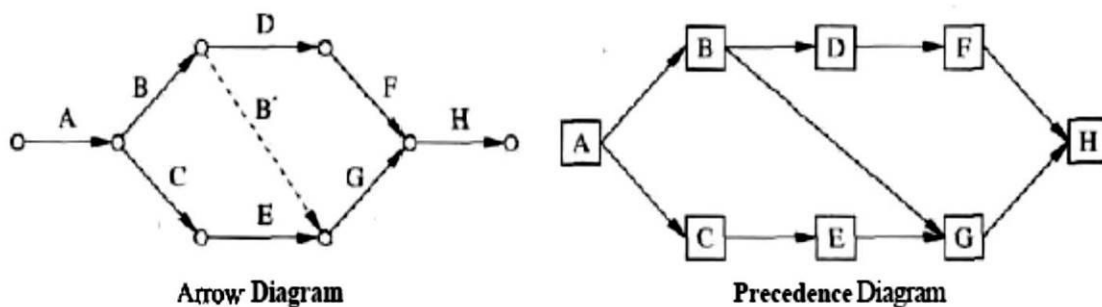
Activity – on – Arrow (A-O-A) Network Planning Technique

Drawing Project Network.

There are two ways that are commonly used to draw a network diagram for a project:

- 1. Activity on Arrow (AOA) representation.**
- 2. Activity on Node (AON) representation.**

Network -A diagram to represent the relationship of activities to complete the project. The network may be drawn as either an "arrow diagram" or a "precedence diagram", see Fig. below:

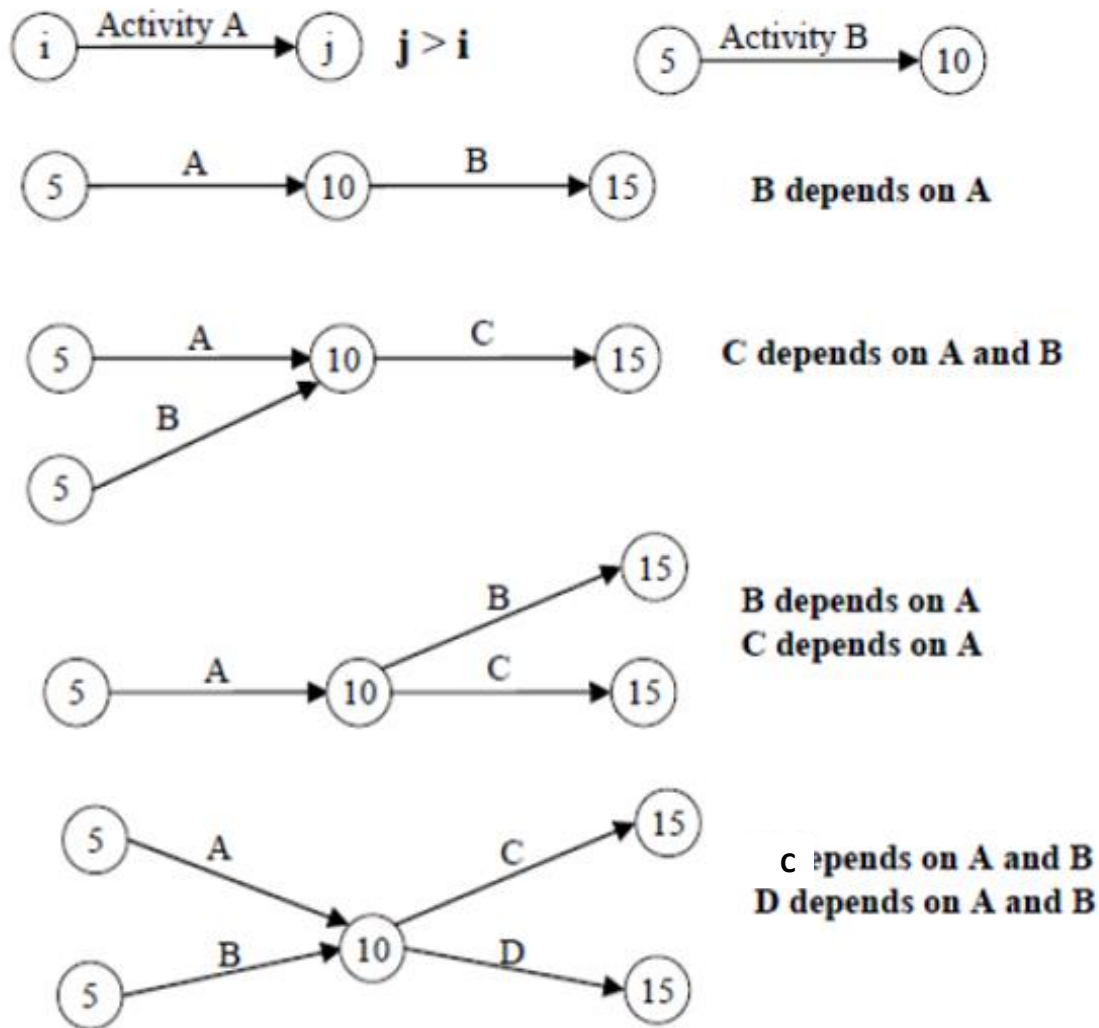


Before drawing the network, it is necessary to ensure that:

- The project has a unified starting and ending point.
- Networks should be continuous (i.e., each activity except the first and the last has both preceding and succeeding activities).

1- Activity on arrow network (AOA)

In this method, the arrows represent activities while the nodes represent the start and the end of an activity (usually named as events) (see Fig. below). The length of the arrow connecting the nodes has no significance. When one activity depends upon another, both appear on the diagram as two arrows having a common node.



The following are some rules that need to be followed when constructing an AOA network diagram:

- Each activity must have a unique $i - j$ numbers, where i (the number at the tail of the arrow) is smaller than j (the number at the head of the arrow).
- It is recommended to have a gap between numbers (i.e., 5, 10, 15, etc.). This will allow for accommodation of missed activities.
- Avoid back arrows.

Dummy Activity -An activity (represented by a dotted line on the arrow network diagram) that indicates that any activity following the dummy cannot be started until the activity or activities preceding the dummy are completed. The dummy does not require any time except the following:

- When more than one arrow leaves the same node and arrives at another node.
- When one activity depends upon two preceding activities and another activity depends only upon one of these two preceding activities as shown in Fig. below.

